

PAMELA QUARTSON

Berkeley, CA | 5107176891 | pamela.quartson@berkeley.edu | [LinkedIn](#) | [GitHub](#)

Professional Profile

Data engineer and full-stack software developer with experience building automated ELT pipelines, integrating third-party APIs, and shipping production-ready data systems across cloud environments. Skilled in Python, SQL, and end-to-end data workflow ownership, from schema design and ingestion through transformation, validation, monitoring, and deployment.

Education

UC Berkeley

May 2026

Master of Information Management and Systems

- **Relevant Coursework:** Principles & Techniques of Data Science & Building Data Products, Introduction to Data Structures and Analytics, Research Design and Applications for Data and Analysis, Applied Generative AI, Applied Natural Language Processing

Graduate Certificate in Applied Data Science

Ashesi University

May 2021

BSc, Computer Science

- **Relevant Coursework:** Data Systems & Machine Learning, Data Science, Data Structures & Algorithms, Algorithm Design & Analysis, Computer Organization and Architecture, Networks and Distributed Computing

Skills

- **Languages:** Python, JavaScript/TypeScript, PHP, SQL, R, Java
- **Frameworks & Tools:** Node.js, NestJS, ExpressJS, ReactJS/Redux, Vue, PostgreSQL, MySQL, Google Firebase, MongoDB, Redis
- **Cloud & Infrastructure:** Google Cloud Platform (GCP), Amazon Web Services (AWS), Docker, Kubernetes, DevOps
- **AI/ML:** RAG, GraphRAG, LangChain, ChromaDB, CrewAI, Amazon Bedrock

Work Experience

UC Berkeley Goldman School of Public Policy | *Lead Data Engineer*

Sep 2025 - Present

- Lead maintenance and expansion of automated Python ELT pipelines for the Voluntary Registry Offsets Database (VROD), integrating multiple third-party registry APIs to ingest, transform, and centralize 10,000+ carbon offset project records into a unified data store.
- Own end-to-end pipeline reliability: designed schema structures, implemented error handling, automated data validation, and built monitoring logic, reducing manual intervention by 30% and improving data accuracy across all registry sources.
- Continuously refine transformation logic and data models to ensure consistent, clean metrics across heterogeneous data sources, supporting market transparency for researchers and analysts.

Street Streams Limited | *Software Engineer (Full-Stack)*

Jul 2022 - Jul 2024

- Designed cloud-native backend infrastructure on GCP (Compute Engine, Cloud SQL, Cloud Run), collaborating with cross-functional teams to build scalable, production-ready systems for the e-commerce platform.
- Built a real-time analytics dashboard aggregating business and user metrics from multiple data sources, reducing leadership decision-making time by 80%, and demonstrating end-to-end ownership from data modeling to visualization.
- Achieved a 25% reduction in API response time through performance optimizations using JavaScript service worker threads and PM2 process manager tool, leading to an improved user experience.
- Provided technical support to merchants and field teams, troubleshooting platform issues and translating user feedback into product improvements.

518Innovations | *Software Engineer (Full-Stack)*

Jun 2021 - Jun 2022

- Collaborated with users to define technical requirements and prioritize features for multiple client products, translating business needs into engineering specifications and managing the enhancement backlog using JIRA
- Built and integrated REST APIs with PHP Laravel, using Postman for testing and Docker for containerization, enabling seamless communication between internal systems and third-party applications.
- Achieved improvement in API response times by 10% through performance optimizations, debugging and efficient caching strategies, contributing to efficient product management and delivery.

Projects

Privacy Regulation GraphRAG System

Jan 2025 - May 2025

- Built a GraphRAG system using LangChain, ChromaDB, and Memgraph to answer complex privacy regulation queries by combining semantic search with knowledge graphs extracted through zero-shot prompting.
- Improved accuracy by 21% and precision by 17% over baseline RAG (0.87 and 0.96 respectively) while maintaining 450ms average retrieval latency.